

Summary Report

Python Assignment 2: Data Structures & Conditional Statements

Objective

The objective of this assignment was to understand and practice Python's core data structures Lists, Dictionaries, and Sets along with conditional statements (if, elif, else). The tasks focused on creating, modifying, accessing, and analyzing data using built-in Python methods.

Tasks Completed

1. List Operations

- Created `age_list` and `name_list`.
- Performed list operations such as:
 - `append()`
 - `insert()`
 - `remove()`
 - `pop()`
 - `extend()`
 - `sort(reverse=True)`
- Accessed elements using indexing and slicing.
- Calculated the **maximum**, **minimum**, and **sum** of the age list.

2. Dictionary Operations

- Created a dictionary to store student names and marks.
- Accessed a student's mark using the key.
- Added a new student (Janani) and updated an existing student's mark.
- Displayed all keys, values, and key-value pairs using:
 - `keys()`
 - `values()`
 - `items()`

3. Set Operations

- Created a set containing duplicate values and observed that duplicates were removed automatically.
- Learned that sets are **unordered** and **do not support indexing**.
- Computed the **union** and **intersection** of two sets using:

4. Conditional Statements

- Developed a program using **if, elif, and else** to categorize performance based on a score (0–10).
- Displayed appropriate feedback for:
 - Above Average
 - Average
 - Below Average

Key Learnings

- Lists are **ordered and mutable**, making them suitable for storing and modifying collections of data.
- Dictionaries store information as **key-value pairs** for fast data retrieval.
- Sets automatically remove duplicate values and support mathematical set operations.
- Conditional statements help make decisions based on different conditions.

Conclusion

This assignment strengthened my understanding of Python fundamentals, including data structures and decision-making statements. I gained hands-on experience with common list, dictionary, and set operations, as well as writing conditional logic to solve real-world programming problems. These concepts provide a strong foundation for developing more advanced Python applications.